Fuel the Engine: Bank Credit and Firm Innovation

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Abstract: This paper builds on the premise of the endogenous growth theory to reveal that bank credit fosters firms' innovation activities, and ultimately, economic growth. I find that credit constraints limit technological progress, which is also the case when credit constraints are instrumented by the existence of information sharing systems. The results hold both in case of adopting existing technologies and in terms of R&D expenditures. The findings also indicate that firms employ bank credit to directly finance innovations. Lastly, R&D spending does translate into innovation outputs, and thereby, foster economic growth.

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1. Introduction

The endogenous growth theory treats firm-level innovation as a very important driver of the economic growth (Romer, 1990; Aghion & Howitt, 1992; Coe & Helpman, 1995; Jones, 1995). Innovation helps firms to expand the technological frontier by investing in the creation and accumulation of knowledge, thereby resulting in the invention of new products, processes or services. But knowledge is non-rival and in many cases also non-excludable, spillover does occur during the process of innovation (Grossman & Helpman, 1991; Acemoglu, Aghion, & Zilibotti, 2006). In relatively backward economies, innovation mostly occurs in a way where firms adopt existing technologies.

Innovation activities are costly. Research and development (R&D) involves persistently large amount of investment. For example, South Korea and Israel spent more than 4 per cent of their GDP on R&D in 2014. R&D expenditures also amount to more than 2 per cent of GDP for emerging economies like Slovenia and Czech Republic.[1] In 2016, R&D spending of the top 1,000 companies worldwide amounts to 680 billion US dollars and equals to 4.2 per cent of their total annual sales on average.[2] The adoption of existing products, processes or services are also costly. Mansfield Schwartz & Wagner (1981) and Levin, Klevorick, Nelson, Winter, Gilbert & Griliches (1987) suggest that imitating a new innovation could cost as much as 50 to 75 per cent of the cost of the original invention. As a result, firms need a considerable amount of resources to finance their innovation activities.

Literature indicates that internal finance is an important financing source for the innovation efforts (Schumpeter, 1942; Himmelberg & Petersen, 1994; Hall, 2002). In addition, Kortum and Lerner (2000) show that venture capital also contributes to the development of patents. The same holds for private equity financing (Lerner, Sorensen & Strömberg, 2011). However, the role of bank credit on innovation progress remains a matter of debate with theoretically ambiguous predictions. On the one hand, bank credit might be ill suited to fund innovation activities that have uncertain returns and generate volatile cash flows (Stiglitz, 1985; Brown, Martinsson & Petersen, 2012). In addition, innovation efforts are very likely to be associated with investment in intangible assets that are more difficult to be collateralised (Carpenter & Petersen, 2002). On the other hand, banks are able to overcome information asymmetries related to innovative firms that are difficult in the public markets (Rajan & Zingales, 2001). Funding innovation with public capital may also disclose sensitive information (Bhattacharya & Ritter, 1983; Maksimovic & Pichler, 2001), or it can be costly to the insiders due to the low tolerance of failures in public markets (Ferreira, Manso & Silva, 2014).

Empirical evidence is also ambiguous. Ayyagari, Demirgüç-Kunt and Maksimovic (2011) and Nanda and Nicholas (2014) both find a positive relationship between the supply of bank credit and innovation activities. In contrast, Hsu, Tian and Xu (2014) find the opposite results. Also, Amore, Schneider and Žaldokas (2013) and Chava, Oettl, and Subramanian (2013) find that banking deregulation fosters innovation while Benfratello, Schiantarelli and Sembenelli (2008) and Alessandrini, Presbitero and Zazzaro (2010) indicate a positive impact of bank concentration on innovation. However, the existing literature does not provide any evidence on whether firms finance their innovation projects directly with bank credit. For example, by

[1] Based on the 2014 World Development Indicators by the World Bank.
[2] Based on the 2016 Global Innovation 1000 by PricewaterhouseCoopers (PwC).

employing the bank loans to finance traditional investment, firms could divert more internal resources or retained earnings to support their innovation activities.

In this paper, I build on the work by Bircan and De Haas (2015) to examine the effect of bank credit on firm-level innovation activities in a cross-country setting. In addition, I also employ information on both the firms' loan characteristics and the financing sources of the working capital to partially fill the gap on the mechanisms between the bank credit and technological progress. Lastly, I build on the premise of endogenous growth theory to reveal that a well-functioning financial system fosters innovation, and ultimately, economic growth.

By utlising a comprehensive survey data on a representative sample of 6,422 firms across 22 emerging European countries, the estimates demonstrate that credit constraints limit firms' innovation activities, which is also the case when credit constraints are instrumented by the existence of either a public credit registry or a private credit bureau. The results hold both in case of adopting existing technologies and in terms of R&D expenditures. Furthermore, the findings show that firms do directly employ bank credit to finance their innovation activities. Lastly, R&D investments are found to be able to translate into real innovations, and thereby, foster economic growth.

The contribution of this paper comes from four aspects. Firstly, I employ a firm-level survey data in a cross-country setting, which allows me to exploit very detailed information on the innovation activities from various aspects, not only R&D and patenting. This allows for a rich analysis of the margins along which credit constraints may affect technological progress. At the same time, the cross-country setting of the paper enables me to examine heterogeneity across countries with different levels of governance effectiveness. Secondly, a key empirical obstacle is the lack of convincing identification strategies to mitigate endogeneity concerns. In this paper, I contribute to the literature by employing the introduction of the information sharing system to instrument the credit constraint of a firm. Thirdly, this paper partially fills the gap by revealing the mechanisms on how bank credit affects technological progress by exploiting the loan terms and the financing sources of a firm. Lastly, this paper adds to the understanding of the endogenous growth of firms in emerging economies where innovation mainly occurs by adopting existing products and processes.

This paper proceeds as follows. Section 2 reviews the data and summary statistics. Section 3 introduces the empirical methodology and lays out the hypotheses. Section 4 describes the results, after which Section 5 concludes.

2. Data and Summary Statistics

In order to empirically identify the impact of credit constraint on firms' innovation activities, I utilise firm-level information to measure both the innovation activities in the last fiscal year and conditions of firms' access to credit. All the firm-level data come from the fifth Business Environment and Enterprise Performance Survey (BEEPS V) conducted jointly by European Bank for Reconstruction and Development (EBRD) and the World Bank during 2013-14. The sample covers 6,422 firms across 22 transition countries from Europe, the Baltic States and the Caucasus.[3] The detailed variable definitions and data sources are listed in Table 1.

[Insert Table 1 here]

2.1. Innovation Activities

Information on firms' innovation activities are obtained from the *Innovation Module*, which is for the first time included in BEEPS V. This new module provides detailed information on the input and output of innovation activities. More specifically, this new module covers four types of innovation, which includes *Product Innovation*, *Process Innovation*, *Organisational Innovation* and *Marketing Innovation*. *Product Innovation* measures if a firm introduced new or significantly improved products or services during the last three years. According to the literature, this type of innovation is the most closely related to patents and would be most likely to be affected by credit constraint. Similarly, *Process Innovation* captures the fact that if a firm had introduced new or significantly improved methods for production or supply of products or services. These two types of innovation are sometimes combined and treated together as "hard" innovation (technological). At the same time, firms also take innovation in a non-technological sense. One type of this "soft" innovation is *Organisational Innovation*, where new or significantly improved organizational or management practices or structures are introduced. The other type is *Marketing Innovation* to measure if a firm introduced new or significantly improved marketing methods.

These four variables measure firms' innovation activities from the extensive margin. Besides, I also construct three measures to capture innovation intensively. *1 Innovation* indicates if a firm introduced any of the four following innovation types: product, process, organisational or marketing innovation. Likewise, *2 Innovation* and *3 Innovation* show if a firm introduced two (three) or more types of innovation. The *Innovation Module*, in addition, includes *R&D Input* to capture whether a firm invested in research and development activities during the last three years. This detailed information enables me to exploit various innovation activities beyond patenting. With these data, I can analyse precisely the relationship between access to credit and firm innovation, from both the intensive and extensive margins.

Summary statistics are reported in Table 2. On average, 43 per cent of the firms engaged in at least one type of innovation. There are 28 (25) per cent of the firms undertook two (three) or more innovation types. Product innovation is the most common type of innovation, with 27 per cent of firms had implemented. Between 20 and 24 per cent of the firms reported an innovation in the other three types. However, there is a significantly smaller portion of firms that invested in R&D, either in-house or contracted with other companies (outsourced).

[Insert Table 2 here]

The data also demonstrate substantial variation across countries in the innovation activities. For example, more than 70 per cent of the firms undertook at least one type of innovation

[3] The 22 countries include: Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Estonia, Georgia, Hungary, Latvia, Lithuania, FYR Macedonia, Moldova, Montenegro, Poland, Romania, Serbia, Slovak Republic, Slovenia and Ukraine.

in Romania. However, in Azerbaijan, the number is less than 8 per cent. Similar patterns can be found in the share of firms with at least two or three innovations. In case of a specific type of innovation, there are also considerable differences. For instance, more than 40 per cent of the firms reported a *Product Innovation* in Croatia, Czech Republic and Romania. By contrast, only less than 15 per cent of the firms had *Product Innovation* in countries like Albania, Azerbaijan, Georgia and Montenegro. Table 3 contains more details about the cross-country variations. The cross-country heterogeneity can also be detected by *R&D Input*. In countries like Albania and Azerbaijan, only less than 5 per cent of firms spent on R&D, while more than 20 per cent of firms had R&D input in Croatia, Czech Republic and Slovenia.

[Insert Table 3 here]

2.2. Access to Credit

To measure firms' access to credit, I follow Popov and Udell (2012) and Beck, Degryse, De Haas and Van Horen (2017) to utilise *Credit Constrained*. I combine three questions in BEEPS V to first distinguish between the firms with and without demand for credit (*Credit Demand*). Among the former group, I then identify the firms that were credit constrained: those that either got rejected by a loan application or were discouraged from applying any loan. This combination allows me to distinguish between firms that do not apply for any loan because they do not need one and those that do not apply because they are discouraged.

I start with question K16: "Did the establishment apply for any loans or lines of credit in the last fiscal year?" For firms that answer "No", I move to question K17, which asks the main reason the establishment did not apply for any line of credit or loan. For firms that answer "Yes", question K18a subsequently asks: "In the last fiscal year, did this establishment apply for any new loans or new credit lines that were rejected?" I group firms that answer both "Yes" to K16 and "No" to K18a as credit unconstrained, and firms are constrained if they answer "Yes" to K18a or answer one of the following "Interest rates are not favorable"; "Collateral requirements are too high"; "Size of loan and maturity are insufficient"; or "Did not think it would be approved" to question K17.

According to the summary statistics in Table 2, virtually half of all sample firms had demand for credit. Among those firms with credit demand, 36 per cent were credit constrained. Thus, bank credit is a vital source of external finance in the sample countries and exerts a major obstacle for the firms that are operating there. Table 3 further reveals substantial variation across the sample countries. The share of firms with a demand for credit ranges from 28 per cent in Latvia to 63 per cent in Romania. While 13 per cent of firms in Czech Republic were credit constrained, more than 60 per cent of firms were constrained in Montenegro.

Figure 1 shows the heatmaps of firms' access to credit and innovation activities. Each point represents a firm in the sample. In Panel A and B, darker (lighter) red indicates a firm with (without) product innovation or R&D input. In Panel C, darker red indicates a firm that needs credit. Similarly, in Panel D, darker red indicates a firm that is credit constrained. As a first sight, a negative correlation between innovation and credit constraint can be detected from these heatmaps.

[Insert Figure 1 here]

The BEEPS V also asks the borrowing firms to disclose the loan characteristics for their most recent line of credit or loan, including the annual nominal *Interest Rate* (in per cent) and the original *Duration* (in months). The borrowing firms also report whether the most recent line of credit or loan required *Collateral* and was dominated in a *Foreign Currency*. The summary in Table 2 indicates that the average interest rate was 10.5 per cent with average duration of 39 months. 80 per cent of all borrowing firms were asked to pledge collateral and 37 per cent of all loans were dominated in a foreign currency like Euro and Dollar.

Information on the financing source of firms' working capital in the last fiscal year was also collected in BEEPS V. Specifically, I construct the proportion (in per cent) of a firm's working capital that was financed from each of the following sources: internal funds or retained earnings (*Internal Financing*); banks (*Bank Financing*); non-bank financial institutions (*Non-bank Financing*); trade credit (*Trade Credit Financing*); money lenders, relatives and friends (*Informal Financing*). *Internal Financing* made up more than 70 per cent of the total amount and was by far the most important source of financing for firms' working capital. In contrast, *Non-bank Financing* and *Informal Financing* together accounted for less than 5 per cent and were not very relevant. Both *Bank Financing* and *Trade Credit Financing* amounted to about 12 per cent, which is quite sizable.

Lastly, I collect information on whether a country has an information sharing system in place, either voluntarily through private credit bureaus (*Private Credit Bureau*) or in the form of mandatory public credit registries (*Public Credit Registry*). Figure 2 discloses that all sample countries have at least one type of information sharing systems.[4]

[Insert Figure 2 here]

2.3. Firm Characteristics

I then construct other firm-level variables using BEEPS V. *Crime Experience* dummy equals to 1 if a firm experienced loss as a result of theft, robbery, vandalism or arson last year and 0 otherwise. *Firm Growth* is another dummy that equals to 1 if a firm expected its annual sales to increase in the next fiscal year. A common set of control variables are further included (i.e. Qi and Ongena 2016; Beck, Degryse, De Haas and Van Horen, 2017). Specifically, I include *Firm Size*, which is classified into small (1-19), median (20-99) and large (100+) firms based on the number of permanent full-time employees. If a firm's annual financial statement is checked and certified by an external auditor, then this firm is classified as an *Audited Firm*. *Female Managed Firm* measures whether the top manager of a firm is female. Firm ownerships are also included, including whether a firm is a *Sole Proprietorship Firm*; a *Publicly Listed Firm*; a *Privatized Firm* from a former state-owned enterprise; and a *Foreign*

[4] The 15 Countries with public credit registries: Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Bulgaria, Czech Republic, Latvia, Lithuania, FYR Macedonia, Montenegro, Romania, Serbia, Slovak Republic and Slovenia. The 18 countries with private credit bureaus: Albania, Armenia, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Estonia, Georgia, Hungary, Lithuania, FYR Macedonia, Moldova, Poland, Romania, Serbia, Slovak Republic, Slovenia and Ukraine.

Firm if more than 50 per cent of a firm's shares are foreign-owned. Detailed summary statistics of these variables are provided in Table 2.

2.4. Country Characteristics

Country-level information on various aspects of the governance is obtained from Worldwide Governance Indicators (WGI) database from the World Bank (Kaufmann, Kraay & Mastruzzi, 2011). *Political Stability* measures the perceptions of the likelihood of the political instability and/or politically-motivated violence, including terrorism. *Rule of Law* captures perceptions of the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence. *Voice and Accountability* captures perceptions of the extent to which a country's citizens are able to participate in selecting their government, as well as freedom of expression, freedom of association, and a free media. All these three measures ranges from approximately -2.5 to 2.5 with higher values indicate better quality of governance.

Figure 3 shows the heatmaps for each of these governance indicators separately. Countries next to Western Europe experience lower likelihood of political instability, better rule of law, as well as more freedom of voice and accountability. This is especially true for Baltic States, Czech Republic, Poland, Slovak Republic and Slovenia. In contrast, low quality of governance occurs in Belarus, Bosnia and Herzegovina, Caucasus countries and Ukraine.

[Insert Figure 3 here]

3. Methodology and Hypotheses

To estimate the impact of firms' access to credit on their innovation activities, I first exploit the following model:

$$Firm\ Innovation_{ics} = \alpha_c + \alpha_s + \beta Credit\ Constrained_{ics} + \gamma X_{ics} + \varepsilon_{ics} \qquad (1)$$

for firm *i* operating in country *c* in industry sector *s*. *Firm Innovation* either measures firms' innovation activities at the extensive margin (product, process, organisational or marketing Innovation) or at the intensive margin (at least 1, 2 or 3 types of innovation). Besides, *Firm Innovation* can also measure firms' spending in R&D. *Credit Constrained* captures whether a firm either got a loan application rejected or was discouraged from applying any loan, which is a measure of firms' access to credit. *X* represents the firm-level control variables including *Firm Size*, *Audited Firm*, *Female Managed Firm*, *Sole Proprietorship Firm*, *Publicly Listed Firm*, *Privatized Firm* and *Foreign Firm*. Country and industry fixed effects are included to control for (un)observed variation at the country and industry level. In addition, these fixed effects also control for country or industry specific innovation spill-overs (Bircan & De Haas, 2015), as well as common credit shocks (Qi and Ongena, 2016). The errors terms are clustered at country-sector level to allow them to be correlated due to country and industry specific unobserved factors.

The main coefficient of interest is β, which identifies the impact of credit constraint on firms' innovation activities. Based on the literature, I hypothesize that if firm *i* operating in country *c* in industry *s* either got rejected in a loan application or was discouraged from applying any credit, this firm *i* would undertake less innovation activities. Hence, a significantly negative β would be expected.

However, in the sample a firm's credit constraint is only observable if the firm needs a loan. This raises the issue of selection bias and the β estimate may be not reliable. To address this issue, I utilise *Crime Experience* as the selection variable that are excluded from Model (1) to identify of the model (Heckman, 1979). *Crime Experience* equals to 1 if a firm experienced losses as a result of theft, robbery, vandalism or arson last year. Economically, for firms that reported losses due to crime, the demand for credit would increase to cover these financial losses, i.e. to repair damaged assets or to replace lost machineries. However, experiencing crime is very unlikely to directly affect firms' access to credit. First-stage Heckman selection model is as below:

$$Credit\ Demand_{ics} = \alpha_c + \alpha_s + \delta Crime\ Experience_{ics} + \theta X_{ics} + \mu_{ics} \qquad (2)$$

for firm *i* operating in country *c* in industry sector *s*. The same sets of control variables *X* are included, as well as country and industry fixed effects. Errors terms are clustered at country-sector level. The hypothesis is that if firm *i* operating in country *c* in industry *s* experienced losses as a result of crime last year, this firm *i* would need more credit. Therefore, I would expect a significantly positive δ. Inverse Mills' ratio is first obtained from Model (2), then it is included in Model (1) to address the selection bias.

The other potential bias of Model (1) comes from the fact that *Credit Constrained* might be endogenous. Specifically, it may be the case that innovation firms are more likely to run into credit constraints. Even if innovation is not funded by bank credit, it may also dry up internal funds or retained earnings available for other activities, which will increases the probability that the firm hits a financial constraint (Bircan & De Haas, 2015). Some unobserved variables may also drive access to credit and innovation simultaneously, which leads to a correlation between them. In order to address the endogeneity issue, I instrument *Credit Constrained* by country-level existence of a *Public Credit Registry* or a *Private Credit Bureau*. Consider the model:

$$Credit\ Constrained_{ics} = \alpha_s + \rho Public\ Credit\ Registry_c + \varphi Private\ Credit\ Bureau_c + \theta X_{ics} + \vartheta_{ics} \qquad (3)$$

for firm *i* operating in country *c* in industry sector *s*. The same sets of control variables *X* are included as well as industry fixed effects. Errors terms are clustered at country-sector level. Theoretical literature reveals that information sharing reduces information asymmetry, and thus promotes firms' access to credit (Padilla & Pagano, 1997; 2000). Empirical works also confirm that information sharing is associated with more lending, fewer defaults and lower interest rates (Jappelli & Pagano, 2002). Thus, I hypothesize that information sharing leads to less credit constraints and expect both significantly negative ρ and φ.

4. Empirical Results

4.1. Main Results

This section starts with Table 4, which provides a vivid illustration that link firms' access to credit with their innovation activities. I focus on three aspects of innovation: the intensive margin which reports the share of firms with at least one, two or three types of innovation; the extensive margin regarding the share of firms with product, process, organizational or marketing innovation; and the share of firms with R&D investment. Firms are categorized in three groups: those with a loan (1,990 firms), those without a loan because they did not need one (3,304 firms), and those without a loan either because they were discouraged from applying or their loan applications were rejected (1,128 firms). Among firms with credit demand, there is a significant difference in the likelihood of innovation activities between those that successfully obtained a loan (50 per cent undertook at least one innovation) and those that failed (40 per cent undertook at least one innovation). Similarly, 32 per cent of all credit unconstrained firms reported a product innovation while only 22 per cent of all credit constrained firms did so. The share of firms that spent in R&D is also significantly lower for credit constrained firms. All of these differences are highly significant under a two-sample t-test with unequal variances. Therefore, at a first sight, innovation occurs mostly in firms that successfully obtained a loan.

[Insert Table 4 here]

I then formally test the relationship between innovation and access to credit by utilising an exact matching strategy. This is to match out both the observed firm characteristics and the (un)observed variables at country and sector level (Ioannidou and Ongena, 2010; Qi and Ongena, 2016). Specifically, each credit constrained firm is matched with all the same credit unconstrained firms based on country, sector and a set of firm characteristics, and the average treatment effect for the treated (ATET) is reported accordingly. The results are reported in Table 5. The significantly positive differences in the innovation activities between credit constrained and unconstrained firms suggest that credit constraints limit innovation. At the intensive margin, the results in columns (1) to (3) suggest that more credit constrained firms are less likely to undertake at least one, two or three different types of innovation activities. Economically, credit constrained firms are 5.4 per cent less likely to take any innovation, and 6.4 (4.4) per cent less likely to carry out two (three) or more innovations.

Column (4) indicates that the likelihood of undertaking a product innovation is significantly lower for credit constrained firms. Economically, credit constrained firms are almost 7 per cent less likely to undertake a product innovation compared to credit unconstrained firms. A similar impact is found in column (5). The probability of carrying out a process innovation is 6 per cent lower for the credit constrained firms. In contrast, credit access has no significant impact on organisational or marketing innovations, as well as on R&D input.

[Insert Table 5 here]

However, the matching results may be biased due to the potential self-selection issue. That is, a firm's credit constraint is only observable if the firm needs a loan. To address this issue, I point to Table 6 where a two-stage Heckman selection equation is applied. The first-stage Heckman selection equation is reported in column (1). The dependent variable is the *Credit Demand* dummy and the selection variable is *Crime Experience*. The same set of firm control variables is also included in the equation, as well as the country and sector fixed effects. As expected, *Crime Experience* is significantly and positively related with the *Credit Demand*. In other words, if a firm experienced losses as a result of theft, robbery, vandalism or arson in the last fiscal year, the demand for credit would increase (by 10 per cent economically) to cover these financial losses.

The second-stage results are presented in columns (2) to (9), where the *Inverse Mills' Ratio* is included to correct for the selection bias. The *Inverse Mills' Ratio* enters significantly at 1 per cent level in most of the regressions, suggesting the existence of selection bias and that estimates obtained through regressions without such a correction may be inconsistent. At the intensive margin, the likelihood of taking any or at least two (three) types of innovation is significantly lower for the credit constrained firms. Economically, these credit constrained firms are 3.4 per cent less likely to carry out any innovation, and 2.8 (2.6) per cent less likely to undertake two (three) or more innovations. Extensively, the credit constraints lower the probability of taking any product innovation by 5 per cent. In contrast, no significant impacts are found on process, organisational or marketing innovations. However, a lack of access to credit indeed decreases firms' R&D investment by 2.4 per cent.

[Insert Table 6 here]

Another issue comes from the fact that *Credit Constrained* might be endogenous. In order to deal with this issue, I further instrument *Credit Constrained* by country-level existence of a *Public Credit Registry* or a *Private Credit Bureau*. Country fixed effects are not included in the regressions. The first-stage 2SLS regression is reported in column (1) of Table 7. I find that in countries with either a *Public Credit Registry* or a *Private Credit Bureau* in place, firms are in general less credit constrained. Or in other words, information sharing fosters credit access in the domestic markets. Economically, the introduction of a *Public Credit Registry* leads to a 12.4 per cent decrease in firms' credit constraints. Similarly, establishing the *Private Credit Bureau* decreases firms' credit constraints by 18 per cent.

Columns (2) to (9) report the second-stage 2SLS results. Similar results are found from both the intensive and extensive margins, as well as on R&D input. But the economic magnitudes are more pronounced in the instrumental variable setting. For instance, credit constrained firms are 78 (44) per cent less likely to take any innovation (at least two types of innovation). At the same time, firms with credit constraints are 77 (43) per cent less likely to engage in a product (organisational) innovation. Lastly, the probability of investing in R&D is 32 per cent lower for financially limited firms. With respect to the validity of the instrument, results in the table show that the Kleibergen-Paap rk Wald F statistic is more than 15, indicating that the instruments are sufficiently strong. Moreover, the Hansen J statistic also shows that the null hypothesis that the two instruments are jointly valid cannot be rejected.

[Insert Table 7 here]

The main results are robust to stricter controls of fixed effects and broader clustering of the error terms. Columns (1) to (3) in Table 8 report the estimation results with stricter control of fixed effects. Specifically, instead of country and sector fixed effects, column (1) controls for country*sector interacted fixed effects. In column (2), country fixed effects are replaced by locality fixed effects to control for (un)observed variables at a more micro level. Column (3) utilise year fixed effects in addition due to the fact that firms were surveyed in different years within BEEPS V. The estimations confirm the main findings, also with similar economic magnitudes. Columns (4) to (6) cluster the error terms at country, sector and locality levels respectively. The significant coefficient estimates show that the baseline results are robust to these alternative clustering strategies.

[Insert Table 8 here]

4.2. Country Heterogeneity

Literature has demonstrated that market structures and government policies play a key role in promoting innovation (Acs & Audretsch, 1998; Baumol, 1996). In the countries with more stable political environment, better enforcement of law and contracts, and more freedom, the magnitude of the rewards that can be earned from innovation would be larger (Anokhin & Schulze, 2009), and makes it more likely that the prospective innovators will be able to capture a larger portion of these proceeds (Baker, Gedajlovic & Lubatkin, 2005; Baumol, 1996). The level of governance therefore helps explain whether firms have more incentive to pursue promising innovations across countries. It follows that better governance should be associated with rising levels of innovation. However, it is not clear whether differences in governance across countries also affect the relationship between firms' access to credit and innovation activities, and if yes, to which direction.

To investigate this issue, I interact *Credit Constrained* with various country-level governance indicators: *Political Stability*, *Rule of Law* and *Voice and Accountability*. The only dependent variable is *Product Innovation*, which is consistently affected by credit constraints. The same set of firm-level control variables is included as well as industry fixed effects. Errors terms are clustered at country-sector level. Results are reported in Table 9. The first three columns exclude country fixed effects to give an indication on the direct impact of governance on the innovation activities while the last three columns include country fixed effects to control for the (un)observed country-level variables.

Columns (1) to (3) again indicate that credit constraints limit firms' innovation activities. To be more specific, the likelihood of undertaking a *Product Innovation* is about 6 to 8 per cent lower if a firm lacks access to credit. As expected according to the literature, in the countries with better political stability (*Political Stability*) and law enforcement (*Rule of Law*), or with more freedom (*Voice and Accountability*), innovation activities are more prosperous. A one-standard deviation increase in *Political Stability* is associated with a 3.6 per cent increase in the likelihood of a product innovation. Similarly, the economic impacts equal to 6.6 (6.1) per cent for *Rule of Law* (*Voice and Accountability*). However, none of these interaction effects are precisely estimated, indicating the non-existence of the heterogeneous impacts. Similar results are found in columns (4) to (6) where country fixed effects are included. As a result, the relationship between credit access and innovation does not vary across countries with regards to governance.

[Insert Table 9 here]

4.3. Financing of Innovation

This paper finds a positive relationship between access to bank credit and firms' innovation, but do firms finance innovation activities directly with bank loans? The literature has been vague about this question. For example, by employing the bank loans to finance traditional investment, firms could divert more internal resources or retained earnings to support their innovation activities (Amore, Schneider & Žaldokas, 2013).

To examine this question, I first focus on firms that have successfully obtained credit in the recent years and explore whether innovative firms experience any differences in: interest rates to pay (*Interest Rate*), original loan duration (*Duration*), if the loan requires collateral (*Collateral*), and more importantly, if the loan is dominated in the foreign currency (*Foreign Currency*). I control for standard firm covariates, as well as country and industry fixed effects. The fixed effects for the year in which the loan was issued are also included.

Results are offered in Table 10. The number of observations varies slightly across different loan characteristics simply due to the missing values in the survey. As shown in Panel A, the firms with a *Product Innovation* experience no significant differences in terms of the *Interest Rate*, *Duration* and *Collateral*. However, innovative firms are 3.5 per cent more likely to take a loan that is dominated in a foreign currency. Similar results are found in Panel B where I only include firms with a loan that was issued last year. One possible explanation is that technological innovation is very likely to be obtained from abroad for emerging economies. Within these countries, innovation mostly involves imitation as firms acquire foreign existing technology and adapt them to the local environment (Acemoglu, Aghion & Zilibotti, 2006; Fu, Pietrobelli & Soete, 2011). As the firms adopt products that were developed elsewhere, there would be a larger demand for foreign currency lending. Therefore, the results in Table 10 indicate that firms directly utilise foreign currency loans to finance innovation activities.

[Insert Table 10 here]

Working capital is the capital of a firm which is used in its day-to-day operations, calculated as the current assets minus the current liabilities (Fazzari & Petersen, 1993). The operating expenses are deducted from the working capital investment, which includes expenses that provide benefits for the current period. According to the Statement of Financial Accounting Standards (SFAS) and the International Accounting Standards (IAS), the costs of innovation activities like R&D expenses should also be expensed in the current period. Therefore, the spending on innovation activities could somehow be imitated by working capital investment. This is especially true that in practice more than 50 per cent of R&D spending is the wages of highly educated employees (Hall, 2002). Therefore, as a second step, I examine the financing source of the firms' working capital investment last year to check if innovative firms employ more bank credit.

Table 11 presents the estimation results. The financing source of a firm's working capital can be classified into five groups: internal funds or retained earnings (*Internal Financing*); banks (*Bank Financing*); non-bank financial institutions (*Non-bank Financing*); trade credit (*Trade Credit Financing*); as well as money lenders, relatives and friends (*Informal Financing*). I find that for a firm with a *Product Innovation*, it employs 3 per cent less internal funds to finance its working capital last year. In contrast, this innovative firm utilises 2.6 per cent more bank credit. No significant differences are found with regards to non-bank finance, trade credit or informal finance. These results provide a good indication that firms do directly employ bank credit to finance their innovation activities.

[Insert Table 11 here]

4.4. Endogenous Growth

There is rich empirical evidence supporting that R&D input drives firms' innovation activities, which in turn, would foster economic growth (Stokey, 1995; Bayoumi, Coe & Helpman, 1999; Shefer & Frenkel, 2005). Building on the premise of the endogenous growth theory (Aghion & Howitt, 1992), I would like to examine whether access to credit impacts firms' innovation input, thereby, output, and finally, economic growth.

As a first step, I would like to examine whether *R&D Input* translates into innovations. R&D plays a critical role in the innovation process. It's essentially an investment in technological knowledge which is then transformed into new products, processes, or services. Specifically, I regress both intensive and extensive measures of innovation on the R&D investment of a firm and control the standard set of firm covariates. The country and sector fixed effects are included.

The estimates in Table 12 imply that for a firm with R&D input, it is more likely to result with more innovation outputs. For example, firms with R&D spending are 41 per cent more likely to end up with one or more types of innovation. The probability of undertaking at least two (three) innovations is 40 (35) per cent higher for these R&D firms. At the extensive margin, firms with R&D expenditures are 37 per cent more likely to employ a *Product Innovation*, 33 per cent higher for a *Process Innovation*, 35 per cent more likely to report an *Organisational Innovation* and 34 per cent higher for a *Marketing Innovation*.

[Insert Table 12 here]

In the second step, I focus on the corresponding effects of innovation output on economic growth. To be more specific, I examine the impact of innovation from both the intensive and extensive margins on the likelihood of a sales growth in the next fiscal year. The results are reported in Table 13. Columns (1) to (4) focus at the intensive margin while columns (5) to (9) center at the extensive margin. Taking the results in column (4) as an example, employing the first type of innovation is associated with a 6.3 per cent increase in the likelihood of a sales growth in the next fiscal year. In addition, the second innovation further increases the likelihood by another 5.5 per cent. Similarly, if I focus on column (9), the results indicate that all the different types of innovation contribute to the growth opportunities of the firm. To be more specific, a product innovation increases the probability of a positive sales growth by 6.6 per cent, while the impact of a process innovation equals 3.5 per cent. Organisational and marketing innovations are both important, which foster sales growth by 5 and 5.4 per cent respectively.

[Insert Table 13 here]

5. Concluding Remarks

Literature specifies that financial intermediaries such as internal finance, venture capital and private equity are important financing sources for innovation activities. However, the role of bank credit on innovation progress remains a matter of debate with theoretically ambiguous predictions and empirically mixed results. This paper builds on the premise of endogenous growth literature to show that bank credit fosters firms' innovation efforts, and ultimately, economic growth. By employing detailed firm-level information in a cross-country setting, I find that credit constraints significantly limit technological progress, both in case of adopting existing technologies and in terms of R&D expenditures.

This firm-level data covers detailed information on innovation activities from various aspects, not only R&D and patenting. This allows for a rich analysis of the margins along which credit constraints may affect technological progress. At the same time, the cross-country setting of the paper enables me to examine the heterogeneity across countries with different levels of governance effectiveness. A key empirical obstacle in such a cross-country research is the lack of convincing identification strategies to mitigate endogeneity concerns. In this paper, I instrument firms' credit constraints by the existence of information sharing systems, either a public credit registry or a private credit bureau. The core results are also valid under this instrumental variable approach, as well as a set of other robustness checks. In addition, this paper partially fills the gap in the literature by revealing the mechanisms on how bank credit affects technological progress, that is, firms do directly employ bank credit to finance their innovation activities. Lastly, in accordance with the endogenous growth theory, I find that R&D expenditures do translate into innovations, and thereby, foster economic growth.

Taken together, this paper emphasizes the importance of bank credit in fostering innovation activities and technological progress. With limited access to credit, firms can be constrained in adopting or developing new products, processes and services, which in turn, would lower their productivity and hamper economic growth. Therefore, these findings are also relevant for the policy makers.

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Table 1. Variable Definitions and Sources

This table includes the variable definitions and sources. BEEPS V is the fifth wave of the Business Environment and Enterprise Performance Survey (BEEPS) conducted in 2013-2014. DB is the Doing Business Database from the World Bank. WGI is the Worldwide Governance Indicators from the World Bank.

Variable	Definitions	Sources
Innovation Activities		
Product Innovation	dummy =1 if a firm introduced new or significantly improved products or services during last three years	BEEPS V
Process Innovation	dummy =1 if a firm introduced new or significantly improved methods for production or supply of products or services during last three years	BEEPS V
Organisational Innovation	dummy =1 if a firm introduced new or significantly improved organizational or management practices or structures during last three years	BEEPS V
Marketing Innovation	dummy =1 if a firm introduced new or significantly improved marketing methods during last three years	BEEPS V
1 Innovation	dummy =1 if a firm introduced introduced one or more of the above innovation types during last three years	BEEPS V
2 Innovation	dummy =1 if a firm introduced introduced two or more of the above innovation types during last three years	BEEPS V
3 Innovation	dummy =1 if a firm introduced introduced three or more of the above innovation types during last three years	BEEPS V
R&D Input	dummy =1 if a firm spent on research and development activities during last three years	BEEPS V
Access to Credit		
Credit Demand	dummy =1 if a firm either applied for a loan or did not apply for a loan for reasons other than no need in the last fiscal year	BEEPS V
Credit Constrained	dummy =1 if a firm either got a loan application rejected or was discouraged from applying any loan in the last fiscal year	BEEPS V
Public Credit Registry	dummy =1 if a country has public credit registry in place	DB
Private Credit Bureau	dummy =1 if a country has private credit bureau in place	DB
Interest Rate	annual nominal interest rate (in per cent) of the most recent line of credit or loan	BEEPS V
Duration	original duration of the most recent line of credit or loan in months	BEEPS V
Collateral	dummy =1 if the most recent line of credit or loan required collateral	BEEPS V
Foreign Currency	dummy =1 if the most recent line of credit or loan was dominated in a foreign currency	BEEPS V
Internal Financing	proportion (in pe rcent) of the firm's working capital that was financed from internal funds or retained earnings in the last fiscal year	BEEPS V
Bank Financing	proportion (in per cent) of the firm's working capital that was financed from banks in the last fiscal year	BEEPS V
Non-bank Financing	proportion (in per cent) of the firm's working capital that was financed from non-bank financial institutions in the last fiscal year	BEEPS V
Trade Credit Financing	proportion (in per cent) of the firm's working capital that was financed from trade credit in the last fiscal year	BEEPS V
Informal Financing	proportion (in per cent) of the firm's working capital that was financed from informal finance in the last fiscal year	BEEPS V
Firm Characteristics		
Crime Experience	dummy =1 if a firm experienced losses as a result of theft, robbery, vandalism or arson in the last fiscal year	BEEPS V
Firm Growth	dummy =1 if a firm expects its annual sales to increase in the next fiscal year	BEEPS V
Firm Size	indicator for firm size based on number of employees in the last fiscal year: small (1-19), median (20-99), large (100+)	BEEPS V
Audited Firm	dummy = 1 if a firm had its annual financial statements checked and certified by an external auditor in the last fiscal year	BEEPS V
Female Managed Firm	dummy = 1 if the top manager of a firm is female	BEEPS V
Sole Proprietorship Firm	dummy = 1 if a firm is a sole proprietorship	BEEPS V
Publicly Listed Firm	dummy = 1 if a firm is publicly listed	BEEPS V
Privatized Firm	dummy = 1 if a firm is privatized from state-owned enterprise	BEEPS V
Foreign Firm	dummy = 1 if more than 50 percent of the firm's shares are foreign owned	BEEPS V
Country Characteristics		
Political Stablility	perceptions of the likelihood of political instability and/or politically-motivated violence, including terrorism	WGI
Rule of Law	perceptions of the extent to which agents have confidence in and abide by the rules of society	WGI
Voice and Accountability	perceptions of the ability to participate in selection, as well as freedom of expression, freedom of association, and a free media	WGI

Table 2. Summary Statistics

This table reports the summary statistics for all the variables. Definitions and sources of the variables are provided in Table 1.

Variable	Obs.	Mean	Std.	Min.	Max.
Innovation Activities					
Product Innovation	6,422	0.27	0.44	0	1
Process Innovation	6,422	0.20	0.40	0	1
Organisational Innovation	6,422	0.22	0.41	0	1
Marketing Innovation	6,422	0.24	0.42	0	1
1 Innovation	6,422	0.43	0.50	0	1
2 Innovation	6,422	0.28	0.45	0	1
3 Innovation	6,422	0.15	0.36	0	1
R&D Input	6,422	0.11	0.31	0	1
Access to Credit					
Credit Demand	6,422	0.49	0.50	0	1
Credit Constrained	3,118	0.36	0.48	0	1
Public Credit Registry	6,422	0.62	0.49	0	1
Private Credit Bureau	6,422	0.86	0.35	0	1
Interest Rate	1,931	10.54	8.30	0	100
Duration	2,246	39	37	1	360
Collateral	2,540	0.80	0.40	0	1
Foreign Currency	2,567	0.37	0.48	0	1
Internal Financing	5,993	72.01	34.25	0	100
Bank Financing	5,993	11.92	22.43	0	100
Non-bank Financing	5,993	0.98	6.75	0	100
Trade Credit Financing	5,993	12.46	24.45	0	100
Informal Financing	5,993	2.63	12.20	0	100
Firm Characteristics					
Crime Experience	6,422	0.17	0.37	0	1
Firm Growth	4,466	0.46	0.50	0	1
Firm Size	6,422	1.53	0.70	1	3
Audited Firm	6,422	0.36	0.48	0	1
Female Managed Firm	6,422	0.21	0.41	0	1
Sole Proprietorship Firm	6,422	0.11	0.31	0	1
Publicly Listed Firm	6,422	0.02	0.14	0	1
Privatized Firm	6,422	0.11	0.31	0	1
Foreign Firm	6,422	0.07	0.26	0	1
Country Characteristics					
Political Stablility	6,422	0.19	0.53	-0.69	1.07
Rule of Law	6,422	-0.03	0.62	-0.92	1.13
Voice and Accountability	6,422	0.15	0.69	-1.55	1.09

Table 3. Access to Bank Credit and Firm Innovation: Across Countries

This table shows the average percentages of innovation activities and access to credit across countries. Table 1 contains all definitions and Table 2 the summary statistics for each included variable.

	Share of firms with at least:			*Share of firms with innovation in:*				R&D Input	Credit	Credit
Country	1 innovation	2 innovation	3 innovation	Product	Process	Organisational	Marketing		Demand	Constrained
Albania	13.82%	8.00%	3.27%	10.18%	4.73%	5.45%	6.55%	1.09%	29.45%	45.68%
Armenia	23.61%	11.15%	5.25%	15.74%	6.56%	7.54%	12.13%	5.25%	54.10%	20.61%
Azerbaijan	7.57%	3.19%	1.20%	1.99%	3.98%	3.59%	2.39%	0.00%	48.21%	55.37%
Belarus	66.04%	47.98%	31.15%	30.84%	37.69%	43.30%	48.60%	11.53%	59.19%	35.79%
Bosnia and Herzegovina	51.74%	34.07%	17.03%	35.65%	24.92%	26.18%	24.92%	14.20%	47.32%	28.00%
Bulgaria	50.22%	27.80%	11.66%	26.91%	17.94%	26.91%	21.52%	9.87%	50.67%	42.48%
Croatia	62.77%	43.38%	26.15%	41.85%	31.38%	34.46%	35.08%	23.69%	50.46%	43.90%
Czech Republic	62.38%	41.90%	19.52%	49.05%	33.81%	23.33%	24.76%	23.33%	36.67%	12.99%
Estonia	36.76%	24.86%	14.59%	25.95%	21.08%	19.46%	16.22%	11.89%	38.38%	16.90%
Georgia	18.55%	12.10%	7.26%	11.69%	10.89%	7.66%	10.48%	5.24%	46.37%	33.04%
Hungary	32.20%	16.53%	11.02%	17.80%	18.64%	11.86%	18.22%	8.47%	47.88%	34.51%
Latvia	33.77%	17.54%	6.58%	20.18%	14.04%	12.28%	13.16%	7.02%	28.07%	42.19%
Lithuania	43.90%	26.34%	16.10%	27.80%	22.93%	23.41%	19.51%	10.24%	49.76%	33.33%
FYR Macedonia	55.26%	42.04%	24.02%	32.43%	22.22%	40.24%	36.64%	12.61%	44.14%	36.73%
Moldova	45.22%	38.24%	25.74%	31.99%	33.82%	30.15%	30.88%	8.82%	44.49%	39.67%
Montenegro	27.17%	14.13%	4.35%	13.04%	9.78%	10.87%	14.13%	11.96%	53.26%	61.22%
Poland	51.83%	32.76%	18.09%	34.47%	22.25%	23.47%	30.56%	9.29%	35.45%	30.34%
Romania	70.23%	51.59%	32.50%	42.95%	38.86%	41.14%	46.59%	13.18%	63.18%	26.26%
Serbia	53.60%	34.89%	17.27%	36.69%	20.86%	23.38%	30.94%	15.47%	59.71%	27.71%
Slovak Republic	33.33%	17.30%	9.70%	21.52%	14.77%	13.92%	14.35%	7.59%	35.02%	27.71%
Slovenia	50.81%	25.20%	13.01%	34.55%	11.79%	21.54%	24.80%	22.76%	49.59%	16.39%
Ukraine	30.79%	15.90%	6.11%	18.32%	12.98%	10.43%	13.99%	6.49%	61.20%	54.47%

Table 4. Access to Bank Credit and Firm Innovation: Univariate Results

This table reports univariate results on the relationship between access to bank credit and firm innovation. *, **, *** indicate significance at the 10%, 5% and 1% level, respectively, for a two-sample t-test of a difference in means with unequal variances. For the t-tests we compare innovation activity among all firms with a loan (top row) with all credit-constrained firms (penultimate row). Table 1 contains all definitions and Table 2 the summary statistics for each included variable.

	Share of firms with at least:			*Share of firms with innovation in:*				*Share of firms input R&D:*	*Observations:*
	1 innovation	2 innovation	3 innovation	Product	Process	Organisational	Marketing		
Firm has a loan	*50.10%****	*33.47%****	*19.60%****	*32.06%****	*25.33%****	*26.48%****	*28.74%****	*14.07%****	*1,990*
Credit Unconstrained	50.10%	33.47%	19.60%	32.06%	25.33%	26.48%	28.74%	14.07%	1,990
Firm has no loan	*39.71%*	*24.89%*	*13.20%*	*24.71%*	*18.10%*	*19.36%*	*21.37%*	*9.07%*	*4,432*
No Credit Demand	39.77%	24.82%	13.26%	25.54%	17.74%	19.25%	20.94%	9.38%	3,304
Credit Constrained	39.54%	25.09%	13.03%	22.25%	19.15%	19.68%	22.61%	8.16%	1,128
Total	*42.93%*	*27.55%*	*15.18%*	*26.99%*	*20.34%*	*21.57%*	*23.65%*	*10.62%*	*6,422*

Table 5. Access to Bank Credit and Firm Innovation: Matching Results

This table shows exact matching to estimate the relationship between access to bank credit and firm innovation. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

	Intersive margin			*Extensive margin*				*R&D Input*
Dependent Variable	1 innovation	2 innovation	3 innovation	Product	Process	Organisational	Marketing	
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Credit Constrained	-0.054**	-0.064***	-0.044**	-0.069***	-0.060***	-0.018	-0.021	-0.014
	[0.048]	[0.009]	[0.028]	[0.003]	[0.010]	[0.415]	[0.357]	[0.375]
Matching Variables								
Country	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Size	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Audited Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Female Managed Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sole Proprietorship Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Publicly Listed Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Privatized Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Foreign Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,690	2,690	2,690	2,690	2,690	2,690	2,690	2,690

Table 6. Access to Bank Credit and Firm Innovation: Heckman Selection Results

This table shows the Heckman selection regressions to estimate the relationship between access to bank credit and firm innovation. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

	First-stage	*Second-stage*							
		Intersive margin			*Extensive margin*				
Dependent Variable	Credit Demand	1 innovation	2 innovation	3 innovation	Product	Process	Organisational	Marketing	*R&D Input*
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Crime Experience	0.100***								
	[0.000]								
Credit Constrained		-0.034*	-0.028**	-0.026**	-0.050***	-0.022	-0.017	-0.011	-0.024*
		[0.055]	[0.049]	[0.005]	[0.002]	[0.139]	[0.293]	[0.503]	[0.067]
Firm Size	0.015	0.053***	0.049***	0.046***	0.026*	0.040***	0.061***	0.055***	0.047***
	[0.184]	[0.000]	[0.000]	[0.001]	[0.090]	[0.006]	[0.000]	[0.000]	[0.000]
Audited Firm	0.047***	0.042	0.007	0.015	0.020	0.016	0.005	0.027	0.026*
	[0.001]	[0.117]	[0.758]	[0.420]	[0.341]	[0.452]	[0.809]	[0.233]	[0.073]
Female Managed Firm	-0.016	-0.020	0.004	0.010	-0.018	-0.009	0.020	0.010	-0.017
	[0.306]	[0.341]	[0.859]	[0.593]	[0.406]	[0.672]	[0.374]	[0.594]	[0.129]
Sole Proprietorship Firm	-0.045**	0.044	0.053*	0.027	0.021	0.039	0.065***	0.011	-0.005
	[0.047]	[0.128]	[0.055]	[0.226]	[0.402]	[0.144]	[0.007]	[0.677]	[0.762]
Publicly Listed Firm	0.009	-0.071	-0.042	-0.043	-0.068	-0.082*	-0.033	0.029	0.003
	[0.835]	[0.272]	[0.422]	[0.244]	[0.127]	[0.066]	[0.536]	[0.604]	[0.938]
Privatized Firm	0.011	-0.012	-0.056**	-0.018	-0.028	-0.005	0.017	-0.080***	-0.016
	[0.521]	[0.670]	[0.039]	[0.382]	[0.286]	[0.863]	[0.464]	[0.000]	[0.494]
Foreign Firm	-0.112**	0.118**	0.159***	0.081**	0.099**	0.084**	0.140***	0.103**	0.059*
	[0.000]	[0.015]	[0.000]	[0.017]	[0.028]	[0.029]	[0.000]	[0.015]	[0.072]
Inverse Mills' Ratio		-1.479***	-2.030***	-1.132***	-1.149***	-1.461***	-1.532***	-1.323***	-0.515
		[0.003]	[0.000]	[0.001]	[0.005]	[0.000]	[0.000]	[0.001]	[0.109]
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.062	0.164	0.146	0.112	0.120	0.118	0.125	0.127	0.094
Observations	6,422	3,118	3,118	3,118	3,118	3,118	3,118	3,118	3,118

Table 7. Access to Bank Credit and Firm Innovation: Instrumental Variable Results

This table shows instrumental variable regressions to estimate the relationship between access to bank credit and firm innovation. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

	First-stage	*Second-stage*							
	Credit	*Intersive margin*			*Extensive margin*				*R&D Input*
Dependent Variable	Constrained	1 innovation	2 innovation	3 innovation	Product	Process	Organisational	Marketing	
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Public Credit Registry	-0.124***								
	[0.000]								
Private Credit Bureau	-0.180***								
	[0.000]								
Credit Constrained		-0.781***	-0.444**	-0.211	-0.771***	-0.109	-0.428**	-0.209	-0.320***
		[0.001]	[0.048]	[0.211]	[0.000]	[0.556]	[0.047]	[0.320]	[0.004]
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Inverse Mills' Ratio	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	No	No	No	No	No	No	No	No	No
Sector Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F statistic		15.772	15.772	15.772	15.772	15.772	15.772	15.772	15.772
Hansen J statistic		0.680	0.316	0.660	0.064	0.527	1.556	0.165	0.010
Observations	3,118	3,118	3,118	3,118	3,118	3,118	3,118	3,118	3,118

Table 8. Access to Bank Credit and Firm Innovation: Robustness Check

This table shows the robustness checks to estimate the relationship between access to bank credit and firm innovation. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	Product Innovation					
Model	(1)	(2)	(3)	(4)	(5)	(6)
Credit Constrained	-0.053***	-0.050**	-0.049***	-0.050**	-0.050**	-0.050***
	[0.002]	[0.045]	[0.002]	[0.011]	[0.040]	[0.004]
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Inverse Mills' Ratio	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Country*sector	Country*sector	Country*sector	Country	Sector	Locality
Country Fixed Effects	No	No	Yes	Yes	Yes	Yes
Locality Fixed Effects	No	Yes	No	No	No	No
Sector Fixed Effects	No	Yes	Yes	Yes	Yes	Yes
Country*Sector Fixed Effects	Yes	No	No	No	No	No
Year Fixed Effects	No	No	Yes	No	No	No
R-squared	0.172	0.485	0.121	0.120	0.120	0.120
Observations	3,118	3,118	3,118	3,118	3,118	3,118

Table 9. Access to Bank Credit and Firm Innovation: Country Heterogeneity

This table shows the regressions to estimate the heterogeneous relationship across countries between access to bank credit and firm innovation. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	Product Innovation					
Model	(1)	(2)	(3)	(4)	(5)	(6)
Credit Constrained	-0.082***	-0.063***	-0.073***	-0.049***	-0.050***	-0.049***
	[0.000]	[0.000]	[0.000]	[0.003]	[0.003]	[0.002]
Credit Constrained*						
Political Stability	0.030			-0.001		
	[0.363]			[0.967]		
Rule of Law		0.024			-0.006	
		[0.353]			[0.800]	
Voice and Accountability			0.004			-0.009
			[0.840]			[0.664]
Political Stability	0.068**					
	[0.028]					
Rule of Law		0.106***				
		[0.001]				
Voice and Accountability			0.088***			
			[0.000]			
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Inverse Mills' Ratio	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	No	No	No	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.058	0.069	0.066	0.120	0.120	0.120
Observations	3,118	3,118	3,118	3,118	3,118	3,118

Table 10. Firm Innovation and Loan Characteristics

This table shows the regressions to estimate the relationship between firm innovation and loan characteristics. Panel A includes all the firms with a loan and Panel B only includes those firms with a loan that was issued in the last fiscal year. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	Interest Rate	Duration	Collateral	Foreign Currency
Panel A: All firms with a loan				
Model	(1)	(2)	(3)	(4)
Product Innovation	0.025	1.565	0.002	0.035**
	[0.945]	[0.282]	[0.877]	[0.027]
Firm Controls	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
R-squared	0.381	0.278	0.097	0.418
Observations	1,913	2,222	2,461	2,476
Panel B: Only firms with a loan issued in the last fiscal year				
Model	(5)	(6)	(7)	(8)
Product Innovation	-0.634	1.073	0.037	0.056*
	[0.344]	[0.625]	[0.151]	[0.076]
Firm Controls	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
R-squared	0.420	0.147	0.144	0.418
Observations	694	790	856	861

Table 11. Firm Innovation and Financing Source of Working Capital

This table shows the regressions to estimate the relationship between firm innovation and the financing source of working capital. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

	Percentage of working capital financed by:				
Dependent Variable	Internal	Bank	Non-bank	Trade Credit	Informal
Model	(1)	(2)	(3)	(4)	(5)
Product Innovation	-2.955***	2.619***	0.237	-0.108	0.207
	[0.007]	[0.001]	[0.305]	[0.884]	[0.503]
Firm Controls	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes	Yes
R-squared	0.079	0.065	0.014	0.089	0.026
Observations	5,993	5,993	5,993	5,993	5,993

Table 12. R&D Input and Firm Innovation

This table shows regressions to estimate the relationship between R&D input and firm innovation. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

	Intersive margin			*Extensive margin*			
Dependent Variable	1 innovation	2 innovation	3 innovation	Product	Process	Organisational	Marketing
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)
R&D Input	0.407***	0.392***	0.348***	0.366***	0.325***	0.345***	0.340***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.220	0.205	0.185	0.173	0.163	0.185	0.172
Observations	6,422	6,422	6,422	6,422	6,422	6,422	6,422

Table 13. Firm Innovation and Expected Growth

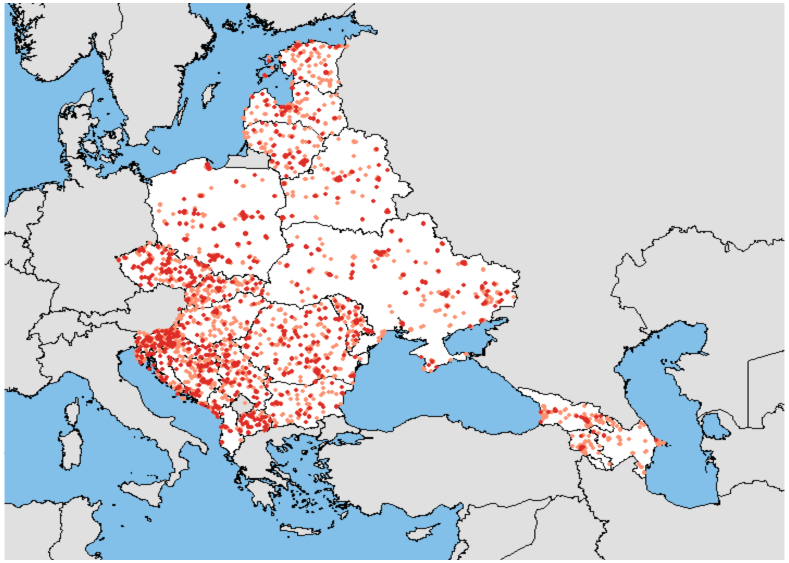
This table shows the regressions to estimate the relationship between firm innovation and expected growth in the next fiscal year. Table 1 contains all definitions and Table 2 the summary statistics for each included variable. Coefficients are listed in the first row, robust country*sector clustered standard errors are reported below in the brackets, and the corresponding significance levels are placed adjacently. *** Significant at 1%, ** significant at 5%, * significant at 10%.

Dependent Variable	Firm Growth								
	Intersive margin				*Extensive margin*				
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
At least 1 Innovation	0.113***			0.063***					
	[0.000]			[0.004]					
At least 2 Innovation		0.128***		0.055**					
		[0.000]		[0.027]					
At least 3 Innovation			0.132***	0.053					
			[0.000]	[0.109]					
Product Innovation					0.110***				0.066***
					[0.000]				[0.004]
Process Innovation						0.105***			0.035*
						[0.000]			[0.074]
Organisational Innovation							0.113***		0.050**
							[0.000]		[0.024]
Marketing Innovation								0.110***	0.054**
								[0.000]	[0.035]
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.097	0.098	0.095	0.101	0.095	0.093	0.094	0.094	0.101
Observations	4,466	4,466	4,466	4,466	4,466	4,466	4,466	4,466	4,466

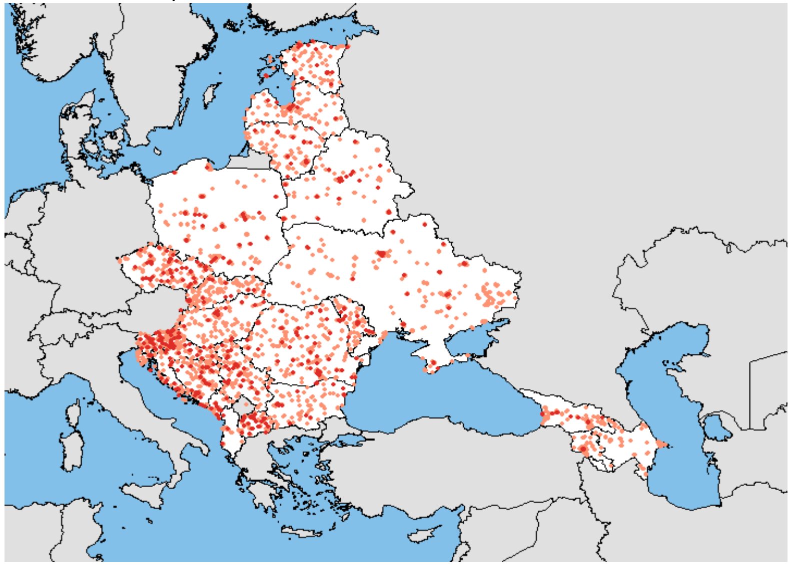
Figure 1. Heatmap of Access to Bank Credit and Innovation

This heatmap reports the access to credit and innovation activities in the last fiscal year for all firms in the sample. In Panel A and B, darker red indicates a firm with product innovation/R&D input and lighter red indicates a firm without product innovation/R&D input. In Panel C, darker red indicates a firm with credit demand and lighter red indicates a firm without credit demand. In Panel D, darker red indicates a firm that is credit constrained and lighter red indicates a firm that is credit unconstrained.

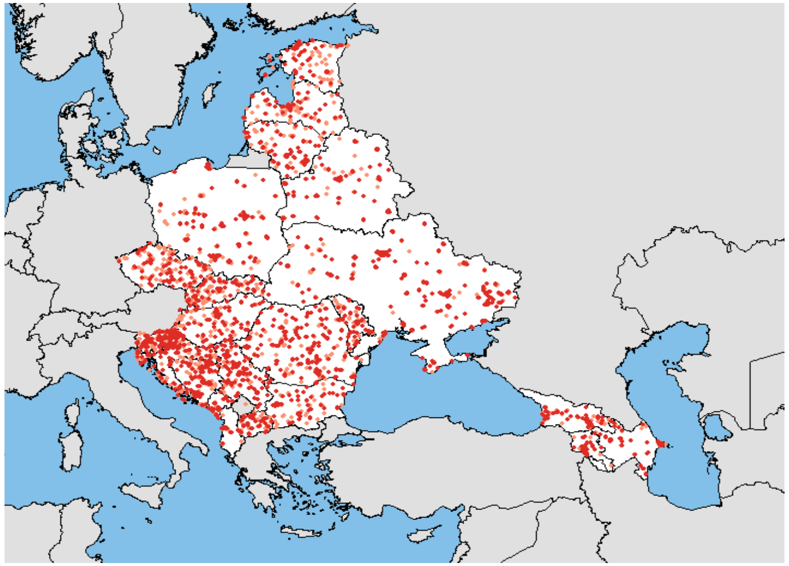
Panel A. Product Innovation



Panel B. R&D Input



Panel C. Credit Demand



Panel D. Credit Constrained

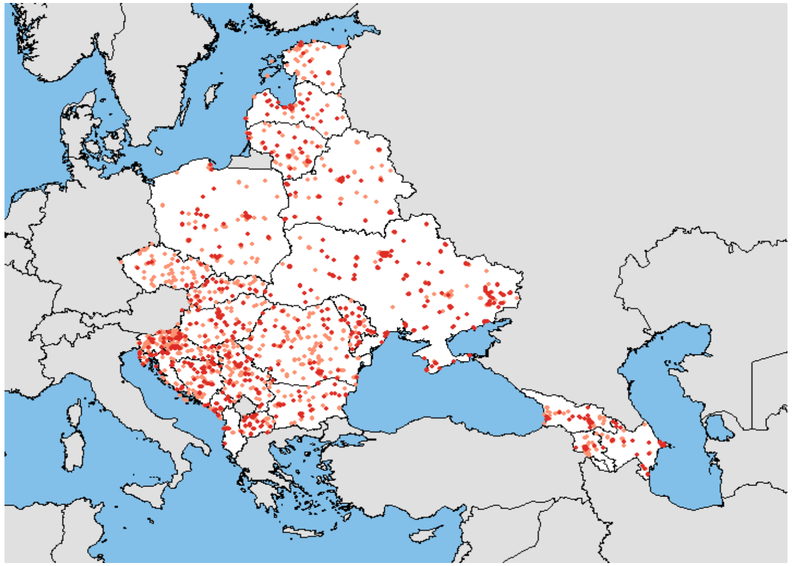


Figure 2. Heatmap of Information Sharing System

This heatmap reports the information sharing system across all the countries in the sample. In Panel A, darker red indicates a country with public credit registry and lighter red indicates a country without public credit registry. In Panel B, darker red indicates a country with private credit bureau and lighter red indicates a country without private credit bureau.

Panel A. Public Credit Registry *Panel B. Private Credit Bureau*

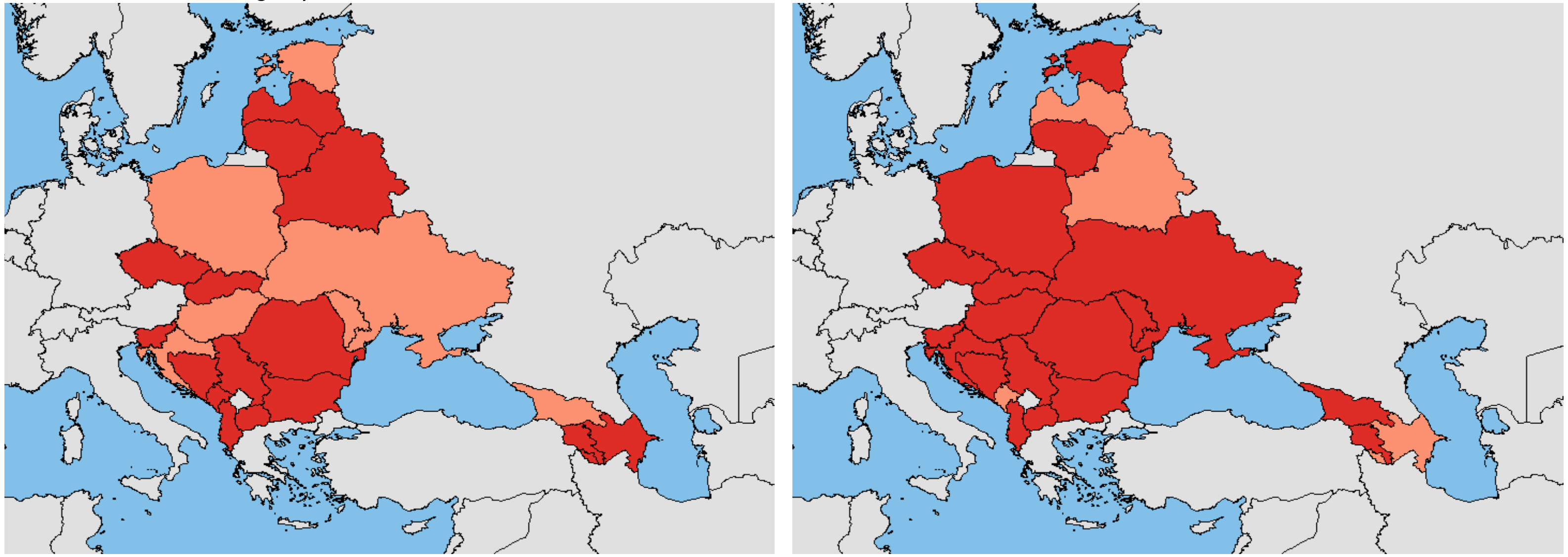
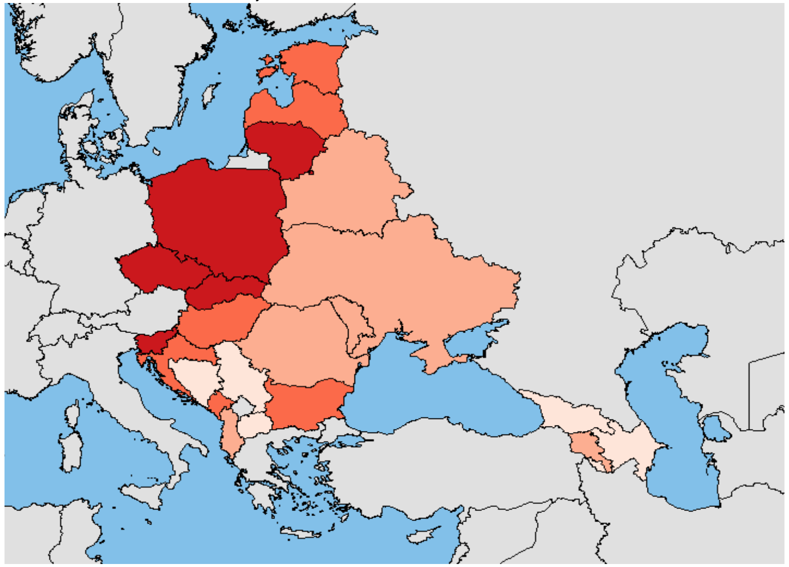


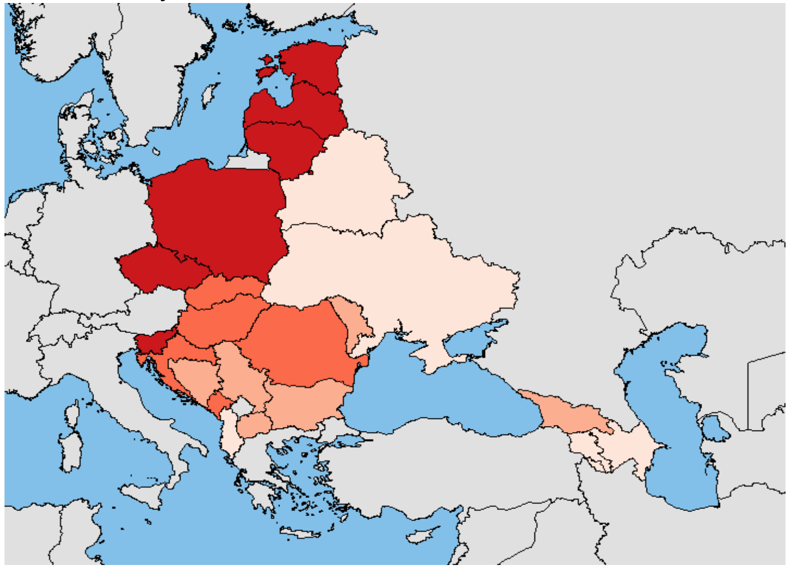
Figure 3. Heatmap of Governance Indicators

This heatmap reports the governance indicators across all the countries in the sample. In Panel A, darker red indicates a country with better political stability. In Panel B, darker red indicates a country with better rule of law. In Panel C, darker red indicates a country with better voice and accountability.

Panel A. Political Stability



Panel B. Rule of Law



Panel C. Voice and Accountability